

Tech Note T25: Are mixed states and random sampling of states equivalent?

Random sampling of trajectories: mixed states, measurements, simulations, and convergence

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Abstract This work investigates the impact of noise channels and random sampling on quantum state measurements. We begin by considering an initial qubit state and analyze how a unitary noise channel affects the state. By employing a twirling procedure over a subset of Pauli operators, we generate two distinct measurement trajectories. We then compare the statistics of these trajectories with those obtained from a mixed state ensemble. Finally, we analyze the moments—including expectation and variance—of both the measurement outcomes and the probability estimates in a setting that mirrors a coin toss experiment. Our key result shows that, despite different constructions (regular mixed states versus trajectory sampling), the average measurement statistics remain equivalent, while their fluctuations capture the intrinsic shot noise.

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25.1 Twirling Variance: A Motivational Simple Example

25.1.1 Variance of Expectation Value of Twirled Channel vs. True Incoherent Equivalent Channel

Example 1 Quantum Channel Representation: The quantum circuits illustrating the noise channel Λ and its twirling are:

$$\rho_0 \xrightarrow{\Lambda := \mathcal{R}_X(\theta)} \rho_n \quad \mapsto \quad \boxed{\Lambda} = \frac{1}{2} \left(\boxed{I} \boxed{\Lambda} \boxed{I} + \boxed{Y} \boxed{\Lambda} \boxed{Y} \right)$$

Definition 1 Initial and Noisy Qubit States: Let $\rho_0 := \frac{1}{2}(I + x_0X + y_0Y + z_0Z)$ be an initial qubit state. Consider a *unitary noise channel* defined by

$$\Lambda(\cdot) := U(\cdot)U^\dagger = \mathcal{R}_X(\theta)\cdot,$$

with $U := R_X(\theta) = \exp(-i\frac{\theta}{2}X) = \cos(\frac{\theta}{2})I - i\sin(\frac{\theta}{2})X$. The noisy state resulting from this channel is given by

$$\rho_n := \Lambda(\rho_0).$$

Definition 2 Twirl Operation: Define the twirl operation using Pauli operators $\{I, Y\}$. Each operator acts via conjugation, producing two trajectory states:

$$\rho_I := \mathcal{I}\Lambda\mathcal{I}(\rho_0), \quad \rho_Y := \mathcal{Y}\Lambda\mathcal{Y}(\rho_0),$$

where $\mathcal{I}(\cdot) := I(\cdot)I$ and $\mathcal{Y}(\cdot) := Y(\cdot)Y$.

Example 2 Measurement Sampling: Consider measuring the observable Z on states ρ_I and ρ_Y . Define random variables $z_I, z_Y \in \{+1, -1\}$ with probabilities

$$\mathbb{P}[z_I = a] = \text{Tr}(|a\rangle\langle a| \rho_I), \quad \mathbb{P}[z_Y = a] = \text{Tr}(|a\rangle\langle a| \rho_Y).$$

The trajectory average is defined as

$$z_t := \frac{1}{2}(z_I + z_Y),$$

with expectation and variance given by

$$\begin{aligned}\mathbb{E}[z_t] &= \frac{1}{2}(\mathbb{E}[z_I] + \mathbb{E}[z_Y]), \\ \text{Var}[z_t] &= \frac{1}{4}(\text{Var}[z_I] + \text{Var}[z_Y]).\end{aligned}$$

Remark 1. The state $\rho_t := \frac{1}{2}(\rho_I + \rho_Y)$ captures the effective twirled state, averaging over the trajectories.

Example 3 Variance Calculations: Explicitly, the variance for different states and measurement settings are:

$$\begin{aligned}\text{Var}[Z \mid \rho_0] &= 1 - z_0^2, & \text{raw state,} \\ \text{Var}[Z \mid \rho_n] &= 1 - (z_0 \cos \theta + y_0 \sin \theta)^2, & \text{noisy,} \\ \text{Var}[Z \mid \rho_t] &= 1 - z_0^2 \cos^2 \theta, & \text{twirl effective,} \\ \text{Var}[Z \mid \text{twirl trajectory}] &= \frac{1}{2}(1 - z_0^2 \cos^2 \theta - y_0^2 \sin^2 \theta) & \text{twirl trajectory.}\end{aligned}$$

Remark 2. Notice the difference in variance due to averaging states versus individual trajectory sampling, highlighting how twirling affects noise statistics.

	idx	Mixed state	Classical variables	$\mathbb{E}[z]$
Raw state	$0, r$	ρ_0	z_r	$1 - z^2$
Noisy state	n	$\rho_n := \Lambda \rho_0$	z_n	$1 - (z \cos[\theta] + y \sin[\theta])^2$
Twirl effective	e	$\rho_e := \frac{1}{2}(\mathcal{I} \Lambda \mathcal{I} + \mathcal{Y} \Lambda \mathcal{Y}) \rho_0$ $= \frac{1}{2}(\rho_I + \rho_Y)$	z_e	$1 - z^2 \cos^2[\theta]$
Twirl trajectory	I, Y, t	$\mathbb{P}[\rho] =$ $\{0.5 : \rho_I := \mathcal{I} \Lambda \mathcal{I} \rho_0, 0.5 : \rho_Y = \mathcal{Y} \Lambda \mathcal{Y} \rho_0\}$	z_I, z_Y, z_t	$1 - z^2 \cos^2[\theta] - y^2 \sin^2[\theta]$

Definition 3 Composite Binary Variable Sampling: Let X be a binary variable sampled from two Bernoulli distributions $\text{Bernoulli}(p_A)$ and $\text{Bernoulli}(p_B)$ with probability $(1 - p)$ and p , respectively. Formally,

$$X \sim \begin{cases} \text{Bernoulli}(p_A), & \text{with probability } 1 - p, \\ \text{Bernoulli}(p_B), & \text{with probability } p, \end{cases}$$

with $0 \leq p, p_A, p_B \leq 1$.

Remark 3. The variance of X is calculated by conditioning on the selected Bernoulli distribution:

$$\text{Var}[X] = (1 - p)p_A(1 - p_A) + pp_B(1 - p_B) + (1 - p)p(p_A - p_B)^2.$$

Remark 4. The quantum circuit above visually emphasizes the sequential application of identity operations and the noise channel, clarifying its implementation context.

25.2 (T25) Are mixed states and random sampling of states equivalent?

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Let us compare a mixed-ensemble state to a quantum trajectory version of it, i.e., one where we randomly sample one of the states in the ensemble for each instance / realization. Let us here only focus on the measurement outcomes. We assume to work over a finite-dimensional Hilbert space $\mathcal{H}$.

25.2.1 Summary and Overview

- **Mixed States vs. Trajectory Sampling:**

- Compare the regular mixed state (a “true probability mixture” at each realization)

$$\rho_R := \sum_{a \in \mathbb{A}} p_a \rho_a$$

with the random trajectory state, which actually chooses only one of the possible elements of the mixtures ρ_a (the K -th) in a given realization,

$$\rho_T[K] := \sum_{a \in \mathbb{A}} [K = a] \rho_a = \rho_K ,$$

where $[K = a]$ denotes the Iverson bracket, which is 1 if K is a and otherwise is zero. The index T stands for trajectory, since we sample from this.

- Demonstrate that the expectation values for measurement outcomes are identical in both constructions.
- Show that the variance in the measurement probability (i.e., shot noise) is given by

$$\text{Var}[P_r] = \frac{p_r(1 - p_r)}{N}.$$

- **Statistical Analysis:**

- Detail the computation of moments of binary outcomes and classical probability estimators.
- Outline the derivation of sample statistics such as:
 - * Sample mean,
 - * Sample variance (with Bessel’s correction), and
 - * Estimators for classical entropies (including the Rényi entropy).

Refs Useful notes on images, see

- PEC Circuit Diagrams for paper circa 2022 - Part 2 - link
- PEC Circuit Diagrams for paper circa 2022 - Part 1 - link

25.2.2 Context

Suppose I define ρ two related ways: regular (R) and trajectory (T) sampled as one of some set of possible variables

$$\begin{aligned}\rho_R &:= \sum_{a \in \mathbb{A}} p_a \rho_a, & \text{regular, mixed state ensemble,} \\ \rho_T[K] &:= \sum_{a \in \mathbb{A}} [K = a] \rho_a, & \text{trajectory, random operator instance,}\end{aligned}$$

over some alphabet $a \in \mathbb{A}$, which represents the labels for each of the trajectories or states in the ensemble, ρ_a is some valid quantum state $\text{Tr}(\rho_a) = 1$, K as a random variable (defined more precisely next), and $[K = a]$ is the Iverson bracket of the logical expression testing if K is equal to a . Notation: We will often omit the explicit dependance of ρ_T on K .

Iverson bracket is 1 if its condition is true, and zero otherwise,

$$[K = a] = \begin{cases} 1, & \text{if } K \text{ is equal to } a, \\ 0, & \text{otherwise.} \end{cases}$$

In this sense, the Iverson bracket can be thought of as just the Kronecker delta $[K = a] \mapsto \delta_{Ka}$. The categorical random variable K is picked from the alphabet set $K \in \mathbb{A}$ with some probability p_a ,

$$\Pr[K = a] = p_a,$$

where $\sum_{a \in \mathbb{A}} p_a = 1$. Note the key properties, where we take the expectation value E over the classical random variable K :

$$\begin{aligned}E_K[[K = a] f(a)] &= \sum_{b \in \mathbb{K}} ([b = a] f(a)) \Pr[K = b] \\ &= \sum_{b \in \mathbb{K}} \delta_{ab} f(a) \Pr[K = b] \\ &= f(a) \Pr[K = a] \\ &= f(a) p_a, \\ E_K[[K = a] [K = b] f(a, b)] &= \sum_{c \in \mathbb{K}} ([c = a] [c = b] f(a, b)) \Pr[K = c] \\ &= \sum_{c \in \mathbb{K}} \delta_{ac} \delta_{bc} f(a, b) \Pr[K = c] \\ &= \delta_{ab} f(a, a) \Pr[K = a] \\ &= \delta_{ab} f(a, a) p_a.\end{aligned}\tag{1}$$

In the right column of expressions, we have made use of the identity $\sum_k \delta_{ik} \delta_{kj} = \delta_{ij}$. One can also ask about more complicated functions, such as expectation of functions that explicitly depend on the trajectory choice, $E_K[f(a, K)] = \sum_{b \in \mathbb{K}} f(a, b) \Pr[K = b] = \sum_{b \in \mathbb{K}} f(a, b) p_b$.

Expectation value directly over states. Averaging over the $E_K[\rho_T[K]] = \sum_{a \in \mathbb{A}} E_K[[K = a]] \rho_a = \sum_{a \in \mathbb{A}} p_a \rho_a = \rho_R$, where we used Eq. (1) in the rightmost equality. The average trajectory density matrix is equal to the ensemble one as expected. This is by construction.

Note that in trajectory simulation you might calculate $\langle \hat{O} \rangle(\rho_a)$. Over different sets of ρ_a

Measurement operators. To sample a measurement outcome, we will sample ρ_R or the random state ρ_T as randomly picked by sampling the random variable K . Let's pick a general quantum measurement of an operator $\hat{M} = \sum_{m \in \mathbb{M}} M(m) \hat{\mu}_m$, where the positive-semidefinite measurement operator is $\hat{\mu}_m \in \text{Pos}(\mathcal{H})$, with $\sum_{m \in \mathbb{M}} \hat{\mu}_m = \hat{I}$, where the system Hilbert space is $\mathcal{H}$, and the outcomes of the measurement are m from the alphabet $\mathbb{M}$, and the operator corresponding value is $M(m)$. For simplicity, we can take $M(m) = m$. The probability to sample a measurement outcome m of the classical random measurement outcome variable M

is $\Pr[M = m] = \langle \mu_m, \rho \rangle = \text{Tr}(\hat{\mu}_m^\dagger \rho) = \text{Tr}(\hat{\mu}_m \rho)$ since $\hat{\mu}_m^\dagger = \hat{\mu}_m$ (every positive semidefinite operator is Hermitian).

$$\begin{array}{c} \{\hat{\mu}_m : m \in M\} \\ \rho \text{---} \boxed{\text{---} \nearrow \text{---}} \text{---} M \sim \Pr[M = m | \rho] = \text{Tr}(\hat{\mu}_m \rho) \end{array}$$

Example: Qubit with measurement operator $\hat{M} = \frac{\hat{I} - \hat{Z}}{2} = 0|0\rangle\langle 0| + 1|1\rangle\langle 1|$, classical outcomes random variable and outcome labels $M, m \in \mathbb{M} = \{0, 1\}$, and measurement operators $\hat{\mu}_0 = |0\rangle\langle 0|$ and $\hat{\mu}_1 = |1\rangle\langle 1|$.

25.2.3 Probability to sample a measurement outcome.

Let's calculate the measurement probability to sample the random classical variable M having the outcome $m \in \mathbb{M}$ for the two case of ρ_R and ρ_T . In the regular case, the probability to sample $M = m$ denoted $p_{m|R}$ is just a number that depends on the state ρ_R . Keep in mind that ρ_T is a random operator, so the probability in the latter case to sample $M = m$ per shot, denoted $p_{m|T}[K]$, will itself become a random variable!

$$\begin{array}{ll} p_{m|R} := \Pr[M = m | \rho_R] & p_{m|T}[K] := \Pr[M = m | \rho_T[K]] \\ = \text{Tr}(\hat{\mu}_m \rho_R) & = \text{Tr}(\hat{\mu}_m \rho_T[K]) \\ = \text{Tr}\left(\hat{\mu}_m \sum_{a \in \mathbb{A}} p_a \rho_a\right) & = \text{Tr}\left(\hat{\mu}_m \sum_{a \in \mathbb{A}} [K = a] \rho_a\right) \\ = \sum_{a \in \mathbb{A}} p_a \text{Tr}(\hat{\mu}_m \rho_a) & = \sum_{a \in \mathbb{A}} [K = a] \text{Tr}(\hat{\mu}_m \rho_a) \\ = \sum_{a \in \mathbb{A}} p_a p_{a,m}, & = \sum_{a \in \mathbb{A}} \textcolor{purple}{[K = a]} p_{a,m}, \end{array} \quad (2)$$

where we have defined $p_{a,m} := \text{Tr}(\hat{\mu}_m \rho_a)$ the probability to measure outcome m given ensemble state a . The only difference is highlighted in purple. Again, note that the expression on the left is a deterministic value and the one on the right is a stochastic one. These expressions dictate everything else there is to know about the problem; i.e., all else follows from here.

Random probability. In the regular case $p_{r|R}$ just looks like a simple nice concatenated sum over the index a . The interpretation here is that the probability to measure $M = m$ is just the sum over the joint probability to get trajectory a and then in that trajectory to measure m . In the trajectory case, the probability $p_{m|T}$ is itself a random variable (though a very special one), since K is a random variable. There are a few tricky points here:

1. **Set of possible outcomes** (support of the discrete random variable) $\Omega[p_{m|T}[K]] = \{p_{a,m} : a \in \mathbb{A}\}$. There are $|\mathbb{A}|$ possible number of outcomes.
2. **Probability** of possible outcome: $\Pr[p_{m|T}[K] = p_{a,m}] = p_a$. This is really what we mean by the trajectory construction. Note that while $p_{m|T}$ is technically a sum over the random variables comprising the random bit $[K = a]$ and some real number, there is only one and exactly one random bit that is 1 and all others are zero for any sample value K , which greatly simplifies the situation here and merely picks out one term in the sum with the corresponding right probability.
3. **Expectation value.** In the regular case, $\mathbb{E}[p_{m|R}] = p_{m|R}$ and $\text{Var}[p_{m|R}] = 0$, and there is only one

possible outcome. In the trajectory case however, using results from Eq. (1),

$$\begin{aligned}
E[p_{m|T}] &= E\left[\sum_{a \in \mathbb{A}} [K = a] p_{a,m}\right] & E[p_{m|T}^2] &= E\left[\sum_{a \in \mathbb{A}} \sum_{b \in \mathbb{A}} [K = a] [K = b] p_{a,m} p_{b,m}\right] \\
&= \sum_{a \in \mathbb{A}} E[[K = a]] p_{a,m} & &= \sum_{a \in \mathbb{A}} \sum_{b \in \mathbb{A}} E[[K = a] [K = b]] p_{a,m} p_{b,m} \\
&= \sum_{a \in \mathbb{A}} p_a p_{a,m} & &= \sum_{a \in \mathbb{A}} \sum_{b \in \mathbb{A}} p_a \delta_{a,b} p_{a,m} p_{a,m} \\
&= p_{m|R} \text{ (note)} & &= \sum_{a \in \mathbb{A}} p_a p_{a,m}^2 \\
& & & \text{(note)} \neq (p_{m|R})^2,
\end{aligned}$$

25.2.4 Details looking at the probability random variable moments (skip)

(You can skip to next subsection. The following is not needed for the latter.)

To compare the regular to the trajectory case:

1. the average expectation values of the probabilities are all the same: i.e., $E[p_{m|R}] = E[p_{m|T}]$
2. the fluctuations (variance) of the probabilities are not the same (since one is deterministic): you can see this in the second moments being different as it comes from $E[p_{m|R}^2] \neq E[p_{m|T}^2]$. But this may not come up as we will see!

Differences and variance. Let us examine the variance and difference closely. We can calculate the variance using $\text{Var}[X] = E[X^2] - E[X]^2$, ie, $\text{Var}[p_{m|T}^2] = E[p_{m|T}^2] - E[p_{m|T}]^2$.

$$\begin{aligned}
E[p_{m|T}] - E[p_{m|R}] &= 0 \\
E[p_{m|T}^2] - E[p_{m|R}^2] &= E[p_{m|T}^2] - E[p_{m|R}]^2 \\
&= E[p_{m|T}^2] - E[p_{m|T}]^2 \\
&= \text{Var}[p_{m|T}] \\
\text{Var}[p_{m|T}] &= \sum_{a \in \mathbb{A}} p_a p_{a,m}^2 - \left(\sum_{a \in \mathbb{A}} p_a p_{a,m}\right)^2 \\
&= \sum_{a \in \mathbb{A}} (p_a p_{a,m}^2 - p_a^2 p_{a,m}^2) - \sum_{a \in \mathbb{A}} \sum_{b \neq a} p_a p_b p_{a,m} p_{b,m} \\
&= \sum_{a \in \mathbb{A}} p_a (1 - p_a) p_{a,m}^2 - \sum_{a \in \mathbb{A}} \sum_{b \neq a} p_a p_b p_{a,m} p_{b,m}.
\end{aligned} \tag{3}$$

The difference comes from the interference of the cross-terms, which are strictly not present in the trajectory case. Note the likeness of the coin toss variance expression $p_a (1 - p_a)$. Note the squares are not linear in p . So it may appear that these are not equivalent. However, we never have access to these variables directly, rather they only determine the statistics of the random variables we will actually sample. So we will have to see how they get affected.

25.2.5 Sampled random variable

Individual outcomes: probability. Suppose we now look at the expression on the random samples outcome M that will give the counts or expectation value of the probability to sample $M = r$. The expression is $[M = r]$ which is one if $M = r$ and zero otherwise. Let's find it's expected value and variance.

The expected value of this expression is, see Eq. (2),

$$\begin{aligned}
E_M [[M = r] | \rho_T] &= \sum_{m \in \mathbb{M}} [m = r] \Pr [M = m | \rho_R] & E_{K,M} [[M = r] | \rho_T] &= E_K \left[\sum_{m \in \mathbb{M}} [m = r] \Pr [M = m | \rho_T] \right] \quad (4) \\
&= \sum_{m \in \mathbb{M}} \delta_{mr} p_{m|R} & &= E_K \left[\sum_{m \in \mathbb{M}} \delta_{mr} \sum_{a \in \mathbb{A}} [K = a] p_{a,m} \right] \\
&= p_{r|R} & &= \sum_{m \in \mathbb{M}} \sum_{a \in \mathbb{A}} \delta_{mr} p_{a,m} E_K [[K = a]] \\
&\vdots & &= \sum_{m \in \mathbb{M}} \sum_{a \in \mathbb{A}} \delta_{mr} p_{a,m} p_a \\
&\vdots & &= \sum_{a \in \mathbb{A}} p_a p_{a,r} \\
&= p_{r|R} , & &= p_{r|R} .
\end{aligned}$$

Both cases boil down the same probability distribution over the outcomes!! Identical! These determine everything there is to know about the distribution.

Functions of the distribution variables. Note that any function we care about (of the random variable M) will be determined by them entirely. (Question: Is it possible to look at some correlations within the data, from two different samples to find differences? No, because the samples are independent.) Using Eq. 4,

$$\begin{aligned}
E_M [f(M) | \rho_R] &= \sum_{m \in \mathbb{M}} f(m) \Pr [M = m | \rho_R] & E_{K,M} [f(M) | \rho_T] &= E_K \left[\sum_{m \in \mathbb{M}} f(m) \Pr [M = m | \rho_T] \right] \\
&\vdots & &= \sum_{m \in \mathbb{M}} f(m) E_K [\Pr [M = m | \rho_T]] \\
&= \sum_{m \in \mathbb{M}} f(m) p_{m|R} , & &= \sum_{m \in \mathbb{M}} f(m) p_{m|R} . \quad (5)
\end{aligned}$$

In other words, the expectation value over the regular ensemble or trajectory here is indistinguishable from the measurement outcomes:

$$E_M [f(M) | \rho_R] = E_{K,M} [f(M) | \rho_T] .$$

However, note that we could think of a more general function $f(M, K)$ which could only be applied in the trajectory case, this is the basis of quasi-distributions (see Sec. 25.2.8).

Example: $f(M) = M$, using Eq. (5),

$$\begin{aligned}
E_{K,M} [f(M) | \rho_T [K]] &= E_M [f(M) | \rho_R] & E_{K,M} [f(M) | \rho_T [K]] &= E_M [f(M) | \rho_R] \\
\text{for } (f(M) = M) &= \sum_{m \in \mathbb{M}} m p_{m|R} & &= \sum_{m \in \mathbb{M}} f(m) p_{m|R} \quad (6) \\
&= \sum_{m \in \mathbb{M}} m \sum_{a \in \mathbb{A}} p_a \text{Tr} [\hat{\mu}_m \rho_a] & &= \text{Tr} \left[\left(\sum_{m \in \mathbb{M}} f(m) \hat{\mu}_m \right) \left(\sum_{a \in \mathbb{A}} p_a \rho_a \right) \right] \\
&= \text{Tr} \left[\left(\sum_{m \in \mathbb{M}} m \hat{\mu}_m \right) \left(\sum_{a \in \mathbb{A}} p_a \rho_a \right) \right] \\
&= \text{Tr} [\hat{M} \rho_R] \\
&= \langle \hat{M} \rangle (\rho_R) \\
&= E_K [\langle \hat{M} \rangle (\rho_T [K])]
\end{aligned}$$

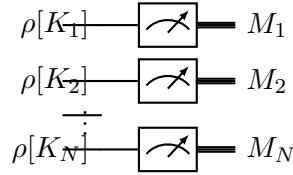
Note that if we used $f(M) = M^2$, we wouldn't see a difference either between the two cases, R and T. Hence, no difference in the variance between the two.

Reminder on functions of operators and the measurement operators

$$\hat{M} = \sum_m M(m) \hat{\mu}_m, \quad \hat{M}^2 = \sum_m \sum_q M(m) M(q) \hat{\mu}_m \hat{\mu}_q.$$

Note that if orthogonal projectors $\hat{\mu}_m \hat{\mu}_q = \hat{\mu}_m \delta_{mq}$ then $\hat{M}^2 = \sum_m [M(m)]^2 \hat{\mu}_m$, and they would be the eigenvalue / eigenvector pair. If such eigen-decomposition, then $f(\hat{M}) = \sum_m f(M(m)) \hat{\mu}_m$, but more generally more complicated.

25.2.6 Multiple instances



Multiple instances. Suppose that we have sampled the random outcome variable M not just once but N times, so we have the set of random variables $\{M_1, \dots, M_N\}$. Note that each of these is independent (which we could get around by using entanglement or other correlations among them and doing bell measurement using a copy of the state ρ but we won't get into that here. There are many quantum advantage tricks to play that way.) Since M_n and $M_{n'}$ for $n \neq n'$ are independent, all the statistics of the total ensemble of outcomes is just governed by those of a single instance. In other words, nothing particularly interesting beyond the standard happens here and there is no difference between the R and T cases. In other words, for a general, non- K dependent function $f(M_1, \dots, M_N)$ over the outcomes

$$\begin{aligned} E_M[f(M_1, \dots, M_N) | \rho_1, \dots, \rho_N] &= \sum_{m_1, \dots, m_N} f(m_1, \dots, m_N) \Pr(M_1 = m_1, \dots, M_N = m_N | \rho_1, \dots, \rho_N) \\ &= \sum_{m_1, \dots, m_N} f(m_1, \dots, m_N) \prod_{n=1}^N \Pr(M_n = m_n | \rho[K_n]), \quad (\text{independent trials}), \end{aligned}$$

where the probability $\Pr(M_n = m_n | \rho)$ is what we calculated in Eq. (4) for a single instance, and we showed it is the same for both the R and T cases. Note that for the random case ρ

Sample statistics. Note that $f(M_1, \dots, M_N)$ can cover just about all key function. Keep in mind that M_n stands for $M_n[K_n]$. Examples include (note S stands for $S(M_1, \dots, M_N)$, etc.):

Statistic	Description / Estimator
$P_r := \frac{1}{N} \sum_{n=1}^N [M_n = r]$	Empirical probability to measure $M = r$
$S := \frac{1}{N} \sum_{n=1}^N M_n$	Sample mean
$V := \frac{1}{N-1} \sum_{n=1}^N \left(M_n - \frac{1}{N} \sum_{n=1}^N M_n \right)^2$	Sample variance (unbiased; Bessel's correction)
$f(M_1, \dots, M_N) = \frac{1}{N} \sum_{n=1}^N f(M_n)$	Average of generic function of the outcome
$H := - \sum_{m \in \mathbb{M}} P_m \log(P_m)$	Classical entropy of the outcomes
$H_\alpha := \frac{1}{1-\alpha} \log \left(\sum_{m \in \mathbb{M}} P_m^\alpha \right)$	Classical Renyi entropy of the outcomes

Sample mean. The sample mean statistic S ,

$$\begin{aligned}
E_{M_1 \dots M_N, K_1 \dots K_N} [S] &= \frac{1}{N} \sum_{n=1}^N E_{M_1 \dots M_n, K_1 \dots K_n} [M_n] \\
&= \frac{1}{N} \sum_{n=1}^N E_{M_n, K_n} \left[M_n \overbrace{[K_n]}^{\text{explicit}} \right] \\
&= \frac{1}{N} \sum_{n=1}^N E_{M, K} [M [K]] \\
&= E_{M, K} [M [K]] \\
&= \langle \hat{M} \rangle (\rho_R) = E_K [\langle \hat{M} \rangle (\rho_T [K])] .
\end{aligned}$$

where we used Eq. (6) for the last equality.

Empirical probability.

$$\begin{aligned}
E_{M_1 \dots M_N, K_1 \dots K_N} [P_r] &= \frac{1}{N} \sum_{n=1}^N E_{M_1 \dots M_n, K_1 \dots K_n} [[M_n = r | \rho_{T,n} [K_n]]] \\
&= \frac{N}{N} E_{M, K} [[M_n = r] | \rho_T [K]] \\
&= p_{r|R} \\
&= \text{Tr} (\hat{\mu}_m \rho_R) \\
&= \langle \hat{\mu}_m \rangle (\rho_R) = E_K [\langle \hat{\mu}_m \rangle (\rho_T [K])]
\end{aligned}$$

using Eq. (4) for the last equality.

Empirical probability square.

$$E_{M_1 \dots M_N, K_1 \dots K_N} [P_r^2] = \frac{1}{N^2} \sum_{n=1}^N \sum_{j=1}^N E_{M_1 \dots M_n, K_1 \dots K_n} [[M_n = r | \rho_{T,n} [K_n]] [M_j = r | \rho_{T,j} [K_j]]] .$$

The argument we can evaluate stand alone. It only depends on 1 or two variables. We should split the two cases when $n = j$ and we refer to the same variable and the case when $n \neq j$ and we refer to different variables:

$$[M_n = r | \rho_{T,n} [K_n]] [M_j = r | \rho_{T,j} [K_j]] = \delta_{nj} [M_n = r | \rho_{T,n} [K_n]] + (1 - \delta_{nj}) [M_n = r | \rho_{T,n} [K_n]] [M_j = r | \rho_{T,j} [K_j]] .$$

Taking the expectation value over each part, using the independence for $n \neq j$, $E_{M_1 \dots M_n, K_1 \dots K_n} [[M_n = r | \rho_{T,n} [K_n]] [M_j = r | \rho_{T,j} [K_j]]] = E_{M_1 \dots M_n, K_1 \dots K_n} [[M_n = r | \rho_{T,n} [K_n]]] E_{M_1 \dots M_n, K_1 \dots K_n} [[M_j = r | \rho_{T,j} [K_j]]]$,

$$\begin{aligned}
&\delta_{nj} E_{M_1 \dots M_n, K_1 \dots K_n} [[M_n = r | \rho_{T,n} [K_n]]] + (1 - \delta_{nj}) E_{M_1 \dots M_n, K_1 \dots K_n} [[M_n = r | \rho_{T,n} [K_n]] [M_j = r | \rho_{T,j} [K_j]]] \\
&= \delta_{nj} p_r + (1 - \delta_{nj}) p_r p_r \\
&= \delta_{nj} p_r (1 - p_r) + p_r^2 .
\end{aligned}$$

Putting it all together,

$$\begin{aligned}
E_{M_1 \dots M_N, K_1 \dots K_N} [P_r^2] &= \frac{1}{N^2} \sum_{n=1}^N \sum_{j=1}^N \delta_{nj} p_r (1 - p_r) + p_r^2 \\
&= p_r^2 + \frac{p_r (1 - p_r)}{N}
\end{aligned}$$

Variance of the probability statistic

$$\begin{aligned}\text{Var}_{M_1 \dots M_N, K_1 \dots K_N} [P_r] &= \text{E}_{M_1 \dots M_N, K_1 \dots K_N} [P_r^2] - \text{E}_{M_1 \dots M_N, K_1 \dots K_N} [P_r]^2 \\ &= \boxed{\frac{p_r (1 - p_r)}{N}}.\end{aligned}$$

This is the key shot noise result! It is independent of R or T case, as follows from the equivalence established in Eq. (4).

Sum of probability estimate squared. Of interest is classical second Renyi entropy argument $\sum_{m \in \mathbb{M}} P_m^2$,

$$\begin{aligned}\text{E}_{M_1 \dots M_N, K_1 \dots K_N} \left[\sum_{m \in \mathbb{M}} P_m^2 \right] &= \sum_{m \in \mathbb{M}} \text{E}_{M_1 \dots M_N, K_1 \dots K_N} [P_m^2] \\ &= \sum_{m \in \mathbb{M}} p_m^2 + \frac{p_m (1 - p_m)}{N},\end{aligned}$$

which in the limit of large number of shots N will be a good estimator for the sum of the probabilities. However, it is a biased estimator.

What of the variance of this non-linear quantity $\text{Var} [P_m^2]$ and $\text{Var} [\sum_{m \in \mathbb{M}} P_m^2]$.

25.2.7 Instance-dependent functions

Suppose we consider a more general function that can depend on the circuit instance K .

25.2.8 Quasi-probability distributions (QPD)

Caveat emptor These pages are a work in progress, inevitably imperfect, incomplete, and surely enriched with typos and unannounced inaccuracies. Sources credited in Bibliography to the best of my ability, though certain omissions certainly remain.

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